

Iteration of linear differential operators

Jake Edmonstone

Matthew Tchouikine

Mentor: Paul Cusson

University of Waterloo, Directed Reading Program

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1. The original problem

Does iterating the derivative infinitely many times give a smooth function whenever it converges?

Asked 4 years, 6 months ago Modified 3 years, 6 months ago Viewed 5k times



I am a graduate student and I've been thinking about this fun but frustrating problem for some time. Let $d = \frac{d}{dx}$, and let $f \in C^\infty(\mathbb{R})$ be such that for every real x ,

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$$g(x) := \lim_{n \rightarrow \infty} d^n f(x)$$



converges. A simple example for such an f would be $ce^x + h(x)$ for any constant c where $h(x)$ converges to 0 everywhere under this iteration (in fact my hunch is that every such f is of this form), eg. $h(x) = e^{x/2}$ or simply a polynomial, of course.



I've been trying to show that g is, in fact, differentiable, and thus is a fixed point of d . Whether this is true would provide many interesting properties from a dynamical systems point of view if one can generalize to arbitrary smooth linear differential operators, although they might be too good to be true.

Perhaps this is a known result? If so I would greatly appreciate a reference. If it has a trivial counterexample I've missed, please let me know. Otherwise, I'll settle with some tricky double limit using tricks such as in [this MSE answer](#), to

Any help is kindly appreciated.

asked Jan 5, 2022 at 6:05



Paul Cusson

1.1 The original question

1. The original problem

Let $f \in C^\infty(\mathbb{R})$, and write $d := \frac{d}{dx}$.

Suppose that for every $x \in \mathbb{R}$ the pointwise limit

$$g(x) := \lim_{n \rightarrow \infty} d^n f(x)$$

exists.

Question. Must g be smooth? Moreover, can we prove $g' = g$?

A direct answer:



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I was able to adapt [the accepted answer](#) to [this MathOverflow post](#) to positively answer the question. The point is that one can squeeze more out of Petrov's Baire category argument if one applies it to the "singular set" of the function, rather than to an interval.

The key step is to establish



Theorem 1. Let $f \in C^\infty(\mathbf{R})$ be such that the quantity $M(x) := \sup_{m \geq 0} |f^{(m)}(x)|$ is finite for all x . Then f is the restriction to $\mathbf{R}$ of an entire function (or equivalently, f is real analytic with an infinite radius of convergence).

The convergence hypothesis actually forces f to be a restriction of an entire function.

answered Jan 6, 2022 at 4:06



Terry Tao

1.2 Tao's analyticity theorem

1. The original problem

Theorem 1

Let $f \in C^\infty(\mathbb{R})$, and suppose that for every $x \in \mathbb{R}$,

$$M(x) := \sup_{m \geq 0} |f^{(m)}(x)| < \infty.$$

Then f is the restriction to $\mathbb{R}$ of an entire function. Equivalently, f is real analytic with infinite radius of convergence.

1.3 Why Tao's theorem applies

1. The original problem

For each fixed $x \in \mathbb{R}$, the sequence

$$(f^{(n)}(x))_{n \geq 0}$$

converges, hence is bounded. Therefore

$$M(x) = \sup_{n \geq 0} |f^{(n)}(x)| < \infty,$$

so Tao's theorem applies immediately: f is the restriction of an entire function.

The easy pointwise observation unlocks global analytic structure.

1.4 One bound for every derivative

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Since f is entire, expand $f^{(k)}$ at 0. Every coefficient is controlled by the *single* number $M(0)$:

$$f^{(k)}(x) = \sum_{n=0}^{\infty} \frac{f^{(n+k)}(0)}{n!} x^n,$$

$$|f^{(k)}(x)| \leq \sum_{n=0}^{\infty} \frac{M(0)}{n!} |x|^n = M(0) \exp(|x|).$$

On every compact interval, this is one integrable bound for *all* derivatives $f^{(k)}$.

1.5 Pass the limit through the integral

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For every n , the Fundamental Theorem of Calculus gives

$$f^{(n)}(a) - f^{(n)}(b) = \int_b^a f^{(n+1)}(x) \, dx.$$

On the compact interval between a and b , the estimate

$$|f^{(n+1)}(x)| \leq M(0) \exp(|x|),$$

is uniform in n . Hence dominated convergence gives

$$g(a) - g(b) = \int_b^a g(x) \, dx.$$

The integral identity also shows that g is continuous.

1.6 The limit satisfies $g' = g$

1. The original problem

Fix b . The integral identity becomes

$$g(a) = g(b) + \int_b^a g(x) \, dx.$$

Since g is continuous, the Fundamental Theorem of Calculus yields

$$g'(a) = g(a).$$

Thus g is smooth, and solving the ODE gives

$$g(x) = C \exp(x)$$

for some $C \in \mathbb{R}$.

1.7 The operator $Lf = af'$

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1.8 The operator $Lf = af' + f$

1. The original problem

1.9 The operator $Lf = af' + bf$

1. The original problem

1.10 A simple zero: $Lf = xf'$

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1.11 A double zero: $Lf = x^2 f'$

1. The original problem

2. Future directions

2.1 The spaces A , B , and C

2. Future directions

2.2 Open Problems

2. Future directions

Questions?